

FOUNDATION_PACKAGES

Foundation Packages (pkg/)

This catalogue describes the pure-Go foundation library that the Helix control plane is built from. There are **129+ packages** under pkg/ (plus 19 private subsystems under internal/).

Every foundation package follows the same engineering contract:

- **Pure Go, standard-library-first.** External dependencies are avoided; where an in-repo primitive is reused it is imported read-only.
- **Deterministic.** Clocks, randomness, and I/O sources are injected (interfaces / seeds / explicit ticks) so tests are reproducible. No `time.Now()` inside logic under test.
- **Mutation-proven (CLAUDE-1).** Each package's tests are designed to *fail* if the specific logic they cover is mutated — green tests over stubs or always-true conditions are forbidden. Items are not marked complete without an independent mutation bite confirming the guard is live.
- **Cross-platform parity (CLAUDE-2).** OS-specific capabilities use real native facilities per platform; there are no Linux-only mocks behind build tags for real-operation features.

Consensus & Coordination

Package	Purpose
voting	Largest-subcluster-wins quorum (3/2 survive+fence, 2/2 both-fence, TTL re-resolve).
failconfirm	Redis-cluster two-phase failure confirmation (PFAIL→FAIL via distinct-reporter majority quorum, OnDead-once).
heartbeatcoalescer	Multi-Raft per-peer heartbeat coalescing, $O(\text{shards} \cdot \text{peers}) \rightarrow O(\text{peers})$, liveness preserved.
leader, lock	Leadership and advisory locking primitives.
splitbrain, splitbrainalert	Split-brain detection and alerting.
phasesgate	Sub-phase dependency-gate validation.
epochresolve	Config-epoch failover slot-ownership conflict resolution (highest-epoch wins, deterministic lexicographic equal-epoch tiebreak, order-independent convergence).

Package	Purpose
multiraft	MultiRaftManager partitioning cluster state into independent per-shard <code>go.etcd.io/raft/v3</code> groups (own leader each) with per-shard Propose routing and an in-process RaftTransport; write throughput scales with shard count instead of hitting the single-leader ceiling, and a shard re-elects on leader loss while others keep committing. Includes a LeaseTracker for CockroachDB-style leaseholder local reads (no Raft round-trip; follower reads route via RaftTransport.SendRead; lease expiry re-routes the fast path). Durability-correct Ready loop: a ShardStorage SetHardState/Append persistence error parks the un-persisted Ready and skips Advance so raft re-presents it (never falsely marking it durably handled) (HXC-917); the persist-error is also surfaced through a public observability seam — a registerable OnPersistError hook + LastPersistError accessor fire with the shard id + error, nil hook = unchanged behavior (HXC-927).
kraft	KRaft-style self-managed Raft metadata quorum (<code>go.etcd.io/raft/v3</code>, no ZooKeeper): replicated CreateTopic/ListTopics + partition-leader assignment via a controller quorum, with controller election and controller-loss failover.
raft	Embedded github.com/hashicorp/raft consensus: a KVFSM replicated key/value state machine (Apply/Snapshot/Restore), a Node wrapper (Apply/Leader/IsLeader/State/Shutdown) and NewInmemCluster for N-node bootstrap over hashicorp/raft's in-memory transport+store. Real 3-node kill-the-leader integration test: a new leader is elected and committed log entries stay intact on survivors, -race clean and non-flaky (HXC-1216). Beyond the in-mem path, three production primitives over the same KVFSM/Node/Cluster contract: (1) a real TCP network transport (<code>tcp.go</code>) — <code>tcpStreamLayer</code> over <code>net.ListenTCP</code> (loopback ephemeral port) wrapped in <code>raft.NewNetworkTransportWithConfig</code>, with <code>NewNetworkCluster(ids...)</code> standing up an N-node cluster whose RequestVote/AppendEntries/InstallSnapshot RPCs travel over genuine kernel sockets (leak-safe partial-failure cleanup; <code>NetTransport/LocalTCPAddr</code> accessors), stores still in-mem (HXC-974); (2) real on-disk persistence (<code>store.go</code>) — <code>NewPersistentNode(id, dataDir)</code> backed by a <code>hashicorp/raft-boltdb/v2 BoltStore</code> (one bbolt file as log+stable store) plus a <code>FileSnapshotStore</code>, so a Shutdown→re-open of the same dir recovers all committed entries from disk WITHOUT caller re-apply (reopen detected → no re-bootstrap; <code>WaitForLeader</code> issues a Barrier so recovery is fully applied before return), in-mem transport (HXC-976); (3) a combined persistent+networked production node (<code>persistnet.go</code>) — <code>NewPersistentNetworkNode / NewPersistentNetworkCluster(ids, dataDirs)</code> composing the real TCP transport AND the real BoltDB+snapshot persistence into one node (restart→recover-from-disk→rejoin over a freshly bound port), reusing <code>openTCPTransport/openPersistentStores</code> so the two halves can't drift (Task I5). For multi-process clusters, <code>NewPersistentNetworkNodeAt(id, dataDir, bindAddr, base, servers)</code> binds a caller-specified fixed TCP address with an explicit server set (first boot bootstraps; reopen recovers-from-disk without re-bootstrap), via the single shared <code>openTCPTransportAt</code> (no TCP-wiring duplication). This primitive backs the <code>helix-raftd</code> daemon (<code>cmd/helix-raftd</code>): one persistent+networked node per OS process with an HTTP admin API (<code>/status</code>, <code>/get</code>, leader-only <code>/put</code> via raft Apply — a follower returns 421 + the real leader address, so no split-brain), proven by a real 3-process E2E

Package	Purpose
	that survives a leader-process SIGKILL and a full-cluster restart-from-disk where every value re-served could only have come from on-disk BoltDB (HXC-982).
raftleader	Real Raft-backed leader-election service adapting raft to a leader-shaped contract: RaftElection (IsLeader/LeaderID/Resign via a real LeadershipTransfer, Apply leader-gated, Campaign/WaitForLeadership, LeadershipChanges sourced from hashicorp/raft's LeaderCh, ErrNotLeader). 3-node test: exactly-one-leader, follower write-gating, kill-leader re-election, committed state survives failover (HXC-953). Closes the gap where leader was not real Raft; leader left untouched.
stonith	Shoot-The-Other-Node-In-The-Head fencing: a FencingAgent interface with IPMI / AWS-EC2 / Azure-ARM / SBD-shared-disk / NoOp drivers and a MultiLevelFencer multi-level fallback, with sink-side fence confirmation (target reachable→unreachable).

Membership & Discovery

Package	Purpose
swim	SWIM gossip membership with phi-accrual failure detection and hierarchical tiers.
discovery (+ discovery/federated, + mdns.go)	Service discovery, including federated cross-cell discovery and zero-config mDNS/DNS-SD LAN discovery on _helix-cluster._tcp (stdlib net + x/net/dnsmesssage): an advertiser publishing TXT records (cellid/nodeid/version/wgpubkey/clusteraddr) and a browser parsing entries into DiscoveredPeer while rejecting any missing/invalid required field. Discovery only — trust requires SPIFFE attestation afterward (live two-node LAN capture tracked as HXC-925).
nattraversal	STUN-based NAT traversal.
ice	ICE connectivity establishment.
cellmesh	Cross-cell mesh networking.
scan	Oracle-SCAN-style stable virtual client endpoint: least-loaded healthy-backend routing with a membership-stable name and injected clock.

Replication & State

Package	Purpose
crdt (+ crdt/merkle)	LWW / OR-Set / G-Counter / PN-Counter / vector-clock CRDTs and Merkle anti-entropy trees.
mvcc	MVCC revision store with a real B-tree index, time-travel reads, watch-from-revision, and compaction.
antientropy	Cassandra-style 3-layer repair: hinted handoff, read repair, Merkle-tree diff (over in-memory replicas).
watchmanager	etcd-style persistent watch manager with synced / unsynced / victim watcher groups.
hlc	Hybrid logical clocks.
offlinesync, checkpoint_merge	Offline delta sync and checkpoint merging.
tieredcache	Hot(memory)/warm(NVMe)/cold(SSD) tiered cache with injected backends + injected clock, idle-demotion + promotion, and per-tier hit stats.
deltacrdt	Standalone delta-state CRDTs (G-counter / PN-counter / observed-remove OR-set / LWW-map) with delta-mutators and a convergent (commutative/associative/idempotent) merge.

Scheduling & Placement

Package	Purpose
scheduler	Omega-model two-level scheduler: ClassAd matching, gang scheduling (all-or-nothing), preemption, optimistic concurrency.
constraints	Pacemaker-style four-type constraint engine (location / colocation / order / stickiness).
preempt	Value-multiplier (Gepetto) preemption with sole-global protection.
priorityqueue	Multifactor composite priority (age + fairshare + size + QoS) with clock-driven aging.
backfill	SLURM-style conservative backfill over an availability timeline.

Package	Purpose
workclaim	SKIP-LOCKED-style optimistic work-claiming (exactly-once under contention).
admissioncontrol	N+K capacity-reserve failover admission control.
budgetcap	Global MaxCostPerHour budget cap on allocations.
qos	QoS-tier routing (real-time / interactive / batch / best-effort).
suitability	HPC-vs-inference workload classifier and placement routing.
ewmarank	Utilization-aware EWMA candidate ranking with sticky-client routing.
fallbackchain	Per-task ordered fallback chains with dedup, attempt cap, and empty-result handling.
providerchain	Ordered multi-tier provider fallback cascade with retrievable/terminal error classification and injected-clock failover SLA.
modelrouter	Strategy-to-default-model resolution (latency / throughput / quality / cost) with a typed unknown-strategy error.
gepetto	HelixGepetto dual-resource local-vs-Chutes arbitration with a monotonic reserve schedule and a strict >0.80 high-water anti-starvation zero-capacity cut-off.
smartrouter	Intelligent ModelRouter (latency / throughput / cost / TEE / balanced) that always excludes unhealthy models, with a typed no-eligible-model error.
llmfailover	Error-classified LLM failover taxonomy (deterministic per-class fallback paths) plus a sandbox scheduling-hint contract.

GPU, Resource & Cost Management

Package	Purpose
pool	Node/instance pool manager with a GPUProvider provisioning seam.
resources	Node resource probe (CPU/mem/disk/net/GPU) extended with accelerator reporting: GPUInfo.TFLOPS + Accelerators{NPUTops, FPGALogicElements, QPUPresent}. Real per-OS probe (darwin Apple-GPU TFLOPS from

Package	Purpose
	system_profiler core count; linux sysfs PCI-identity catalog) with an operator override-label fallback for non-self-probeable NPU/FPGA/QPU (HXC-1300).
local	Local GPU TCO model (amortized capital + power, utilization-scaled).
costsched, latencysched	Cost-aware and latency-aware schedulers.
healthmonitor	Edge-triggered health transitions with auto-failover / recovery callbacks.
healthprobe	Startup / readiness / liveness probe tiers with a startup grace period.
gpuattest	Device attestation crypto: challenge/response + fingerprint, seeded-matmul proof-of-GPU-work, O(1) Merkle spot-check, device-sealed AES-GCM/HKDF.
attestadmit	Attestation-gated scheduler admission predicate (uses gpuattest).
quantization	Per-tier model-variant selection (memory-fit + backend-support gates).
capability, deviceprofile, device, devicemap	Capability negotiation and device profiling.
devicecatalog	Machine-readable 64-device taxonomy with case-insensitive substring Lookup resolving a discovered CPU model (e.g. Rockchip RK3588) to its tier / trust-level / compute-class.
tierdef, tiersec	Tier definitions and tier security.
burst, cloudspot, marketplace	Burst-to-cloud autoscaling, spot pricing, and compute marketplace. cloudspot includes real AWS IMDSv2 (token PUT→authenticated GET instance-action), Azure scheduled-events, and GCP preemption interruption pollers (genuine token-dance wire protocol, http test-proven) driving drain→checkpoint→upload via an injectable sink (HXC-1328; live cloud HXC-1617).
revenueopt	RevenueOptimizer.OptimizeAllocation greedy GPU→marketplace assignment maximising expected revenue, with a TEE→Chutes revenue bias.
carbonsched	Carbon-aware job placement: selects the lowest-carbon-intensity latency-eligible region (deterministic name tiebreak) with per-job kWh / gCO2 metering.
workloadrouter	UnifiedManager.RouteWorkload: concurrent per-marketplace price probes + a weighted composite score (inverse price/latency, health) with a TEE multiplier re-ranking confidential offers when the workload requires a TEE.
economics	RewardDistributor multi-token distribution with treasury/reinvest splits and a conservation invariant, plus GetParticipantROI (ROI% + break-even days, typed unknown-participant error).

Package	Purpose
bursthysteresis	BurstController with a MONITOR→SPILL→RECOVER two-threshold hysteresis dead-band.
gpucatalog	SUPPORTED_GPUS compute-multiplier catalog, LookupMultiplier, and attestation-aware Score (attested scores rank above un-attested).
gputopo	Topology-aware NUMA/NVLink GPU placement scoring that prefers NVLink-connected GPU pairs and falls back to PCIe.
balancemonitor	USD account-balance monitor with a strict low-balance floor warning, over HTTP with Authorization: Bearer.
deviceplugin	Nomad/K8s-style gRPC device-plugin / GRES fingerprinting framework: device plugins register over real gRPC reporting model / memory / driver / PCIe / capabilities / health / utilization; the registry parses GRES descriptors (gpu:rtx3080:1,memory:10Gi), atomically applies fingerprints, allocates by request, and rejects oversubscription beyond reported device count.
gpu	ProviderAdapter registration hooks: a thread-safe registry (Register/List/Get/Lookup, duplicate-name guarded, race-free under -race) exposing a pool-facing provider listing so a new external provider adapter plugs into the GPU layer (HXC-1533; pool consumption HXC-1615).
benchmark	Micro-benchmarker producing a real, repeatable normalized on-host CPU score (+ typed GPU TFLOPS / NPU TOPS seams) for tier assignment / scheduling hints; repeatability asserted within a coefficient-of-variation tolerance band (HXC-1308).
tierdetect	Pre-provision host-capability gate: real cross-platform detection (darwin sysctl/arch + linux /dev/kvm,/proc,binfmt behind one build-tagged interface) returning a typed MissingCapabilityError when a requested tier's needs are unmet (HXC-1230).

Federation & Multi-cluster

Package	Purpose
federation(+ federation/suspicion)	Federation membership and suspicion.
fedtopology	Federation topology patterns.
fedtrust	Cross-cluster federated-trust admission.
configsync	CRDT (LWW) config synchronisation.
residency	Data-residency admission evaluation.
raftprofile	etcd Raft WAN tuning profiles.
spiffefed, doublecrypt	SPIFFE federation (mTLS + x509) and double-encryption.
internal/federation	Karmada PropagationPolicy + OverridePolicy engine: CRs modeled

Package	Purpose
gitops	as Go structs → YAML (no client-go dep), constraint-aware (data-locality/latency/cost/compliance) two-level cell selection, and failover reselect that excludes a downed/NotReady cell; deterministic failover proven via in-process FakeHub (live <60s Karmada capture tracked as HXC-924). ArgoCD ApplicationSet federation client over an injectable HTTP API: matrix generator (one Application per cell), syncPolicy prune+selfHeal, RollingSync canary→tier-2 (50%)→tier-1 (25%) progressive rollout, and drift/prune detection; proven against an httptest ArgoCD (live UI capture tracked as HXC-922).

Messaging, Flow & Routing

Package	Purpose
flowcontrol	Kubernetes API Priority & Fairness (FlowSchema → PriorityLevel → fair queues).
workqueue	Rate-limited work queue with exponential backoff and dedup.
ratelimit, backoff, retry	Rate limiting, exponential backoff, retry.
idempotent	Exactly-once producer via producer-id + sequence.
rebalance	Kafka cooperative-sticky incremental partition rebalancing.
informer	Informer-style list-watch local cache with indexers and resync.
redundantexec	BOINC-style redundant-execution trust scorer.
hashslot	CRC16 16,384 hash-slot router with MOVED/ASK in-flight migration redirection.
slotmigration	Atomic live-session slot migration FSM (PREPARE → TRANSFER → COMMIT/ABORT).
fiber, pubsub, events, serde	Framed transport, pub/sub, events, serialization.
events (Avro)	Stdlib-only Avro-subset codec for domain events: flat primitive records, spec-faithful binary encoding (zig-zag

Package	Purpose
	varint, LE-IEEE-754 double), single-object framing (0xC3 0x01 + CRC-64-AVRO fingerprint) so payloads are self-describing, a SchemaRegistry for schema-VALIDATED routing (unknown fingerprint rejected), and writer→reader schema RESOLUTION enabling backward+forward-compatible evolution (defaulted fields fill reader-only, writer-only fields skip). NodeEventSchemaV1/V2 evolution round-trips both directions (HXC-1119). Wired into the live NATSBackend wire path (WireAvro mode: NodeEvent published as an Avro single-object payload, subscriber decodes via the registry; JSON default preserved), proven against a real NATS broker via brokertest.StartNATS asserting the 0xC3 0x01 marker on the wire (HXC-1626).

Edge

Package	Purpose
edge	Edge restriction, schedule rules, trust, and work units.
edgeregistry	Multi-tier (T3–T8) edge device registration with tier/trust/capability labels.
edgeverify	Redundant edge-output verification against a trusted anchor via SHA-256 checksum.
edgefusion	Edge data fusion.
edgeheartbeat	Edge heartbeat carrying battery%/charging/CPU-temp/network-type with a churn-tolerant collector that marks a device offline past a configurable timeout; real per-OS TelemetrySource (darwin pmset/sysctl, linux /sys+/proc/net) behind one interface, injected clock, validated over a real loopback transport (HXC-1185).
powergater	Charging-gated work-acceptance guard: CanAcceptWork() (bool, reason) returning not_charging/battery_low_X%/daytime/cpu_hot_XC from a real per-OS

Package	Purpose
	PowerSource (darwin pmset, linux / sys/class/power_supply+thermal) behind one interface (HXC-1175).

Security & Verification

Package	Purpose
crypto, jwt	Cryptographic primitives and JWT auth.
modelintegrity	HF-cache verification gate (SHA-256 + size).
gravaladmit	HMAC-SHA256 GraVal challenge-response provider admission gate with single-use nonce and a graval.verified label.
gravalverify	GraVal GPU attestation with a VRAM-ratio ≥ 0.95 gate and a concurrent, race-clean BatchVerify pass-rate KPI.
fsresidency	Filesystem residency challenge: per-byte-range ReadAt + SHA-256 verification proving a model file is really present on disk; truncated / missing / mismatch all fail.
exportcontrol	Export-control / country-code KYC gate on node onboarding: a controlled GPU (H100/A100) from a Tier-3 (embargoed) or unknown jurisdiction is rejected with ErrExportDenied; Tier-1 is allowed.
hybridkex	Hybrid post-quantum key exchange combining X25519 (crypto/ecdh) + ML-KEM-768 (crypto/mlkem) shared secrets via HKDF-SHA256 (negotiated group X25519MLKEM768); both parties derive an identical 32-byte key, and a tampered ML-KEM ciphertext yields a non-matching key.
e2eebench	MeasureHandshakeLatency benchmarks the hybridkex full handshake (median + P95 over many iterations), proving median $< 1\text{ms}$ on host and enforcing per-iteration key agreement (ErrKeyDisagreement).
compliancecdoc	EU AI Act compliance-documentation pipeline: generates a model card + technical-documentation artifact from attestation-log records with a SHA-256 provenance hash bound to the record

Package	Purpose
	content; empty logs → ErrNoAttestations.
gpuattest (multigpu.go)	Multi-GPU node enumeration: an N-GPU node descriptor yields N distinct, independently-verifiable per-device attestation proofs keyed by GPU UUID; rejects duplicate/empty descriptors.
redundantexec, attestadmit, gpuattest	See above — redundancy trust, attestation admission, attestation crypto.

Testing, Simulation & Quality

Package	Purpose
testing/dst (+ BUGGIFY, chaos, turmoil)	FoundationDB-style deterministic simulation testing, BUGGIFY fault macros, network simulation.
dst	Standalone FoundationDB-style deterministic simulation harness: single-threaded seeded-RNG event loop with injectable network (drop/delay/reorder) / disk (fault/stall) / logical clock; same seed yields a byte-for-byte identical trace and exact failure replay (independent of testing/dst).
porcupine	Self-contained WGL linearizability checker + concurrent history recorder: CheckOperations(model, history) returns linearizable PASS / FAIL with the violating operation; verifies distributed-op histories under fault injection.
timefault	Clock-fault injectors (skew / freeze / monotonic-violation) + skew detector for split-brain avoidance.
chaosexp	Chaos experiment suite with delivery-level steady-state.
fmea	FMEA catalogue + RPN validator.
phasegate, phase7matrix	Phase dependency gates and the Phase-7 gap-matrix verifier.
qualitygate, covgate, stats, sandbox	Quality gates, coverage gates, statistics (Welch t-test), sandboxing.

External Provider Integration

Package	Purpose
chutes	Chutes inference client + attestation + E2EE envelope/stream, plus ChutesMinerConfig / ValidatorConfig deployment descriptors with Validate() invariants (non-empty IDs/hotkeys, GPUCount>0, non-negative cost, TEE consistency).
provider/chutes	OpenAI-compatible /v1/chat/completions provider (configurable base URL + cpk_Bearer key, ChatCompletionRequest/Response/Message/Usage) with HTTP-429 retry + ctx-interruptible backoff that honors Retry-After, and typed non-retryable errors.
provider/runpod	RunPodProvider serverless GPU adapter (implements pool.GPUProvider) with a warm-pool fast path (pre-warmed instances served before cold provisioning) over an injectable client. Provision releases the mutex across the ColdProvision RPC (reserve-slot → unlock → RPC → relock → record/rollback) so concurrent cold provisions overlap while a reserved counter keeps capacity correct; EndpointOf(id) exposes the reachable worker endpoint (HXC-919).
provider/aws	AWSProvider EC2 Spot adapter (implements pool.GPUProvider) over an injectable EC2-client interface (no aws-sdk dependency): GPU model→instance-type selection (p5/p4de/g5), Spot + tagging, and a TOCTOU-safe capacity gate (launched+reserved).
provider/ionet	IONetProvider over the io.net REST API (cloud.io.net/api/v1, injectable base URL + *http.Client + bearer; implements pool.GPUProvider): DeployCluster (gpu_type/count/region/image/command/env) returns the backend-minted cluster run-UUID (parsed, never fabricated), HealthCheck derives healthy state on the active cluster, and a concurrency-safe reserve-under-lock capacity gate; closure

Package	Purpose
	proven against a net/http/httptest io.net backend.
chutesaccount	Chutes API client over HTTP (Authorization: Bearer): model-list (TEE / price fields) and /users/me account-balance retrieval.
llmadapter	Claude Messages / OpenAI Responses-API request and response shape adapters to OpenAI chat-completions, preserving tool-call order.
marketplaceadapter	MarketplaceAdapter interface (Name/GetCurrentPricing/SubmitWork) with a Name()-keyed dispatch Registry, an HTTP-backed ChutesAdapter (non-2xx rejected), and an AkashAdapter (Cosmos/AKT reverse-auction pricing, SDL submission, provider-reputation gate) over an injectable client; routes work to the named marketplace. buildSDL YAML-double-quote-encodes caller-controlled WorkSpec fields (yamlQuote) so a crafted value cannot inject manifest structure, and the reputation gate clamps a non-positive MinReputation to a 0.5 safe floor so a zero-value AkashAdapter{} cannot admit everyone (HXC-918).

Observability

Package	Purpose
metrics	Metrics collection (Prometheus-text NodeCollector); plus TierMetrics (tiermetrics.go) exposing gpu_tier_utilization, gpu_cost_per_hour, and provider_health series, and EarningsMetrics (earningsmetrics.go) exposing tao_earnings_total, graval_attestation_status, token_throughput, and gpu_utilization series — all with deterministic ordering and concurrent-safe setters.
tracing	W3C distributed tracing.
health	Health aggregation; includes miner-api (HTTP liveness) and GraVal bootstrap

Package	Purpose
	DaemonSet attestation-readiness dependency checks wired into the rollup, which names the specific failing check on a healthy→unhealthy transition.
grafanadash	Grafana dashboard generation + validation.
log	Structured logging.

This catalogue is maintained as the foundation library grows. For the authoritative list run `ls pkg/`. For the work-item registry that tracks package delivery, see [HXC_REGISTRY.md](#).